

Question 1 – Rain and Wet Ground

1. Propositional Logic

Let:

- **R** = It is raining.
- **W** = The ground is wet.

Knowledge Base:

1. If it rains, then the ground becomes wet:
 $R \rightarrow W$
2. It is raining:
R

Goal:

W

2. Convert to CNF

The implication:

$R \rightarrow W$

is equivalent to:

$\neg R \vee W$

So the clauses are:

- **C1: $\neg R \vee W$**
- **C2: R**

3. Negate the Goal

Goal = **W**

Negated goal:

$\neg W$

Therefore:

- **C3: $\neg W$**

4. Resolution Steps

Resolve **C1** and **C2**:

C1: $\neg R \vee W$

C2: R

Resolving R and $\neg R$:

C4: W

Now resolve **C4** with the negated goal:

C4: W

C3: $\neg W$

Therefore:

$\square$ (**Empty Clause**)

Resolution Tree

$$\begin{array}{c} \neg R \vee W \quad R \\ \backslash \quad / \\ W \\ | \\ \neg W \\ | \\ \square \end{array}$$

Conclusion

Since the **Empty Clause** ($\square$) is derived, the goal is proved.

Therefore, the ground is wet.

Question 2 – Student Assignment Submission

1. Propositional Logic

Let:

- S = Rahul submitted the assignment.
- M = Rahul receives internal marks.

Knowledge Base:

1. If Rahul submits the assignment, then Rahul receives internal marks:
 $S \rightarrow M$
2. Rahul submitted the assignment:
 S

Goal:

M

2. CNF

Convert:

$S \rightarrow M$

to:

$\neg S \vee M$

Clauses:

- $C1: \neg S \vee M$
- $C2: S$

3. Negate the Goal

Goal = M

Negated goal:

$\neg M$

- $C3: \neg M$

4. Resolution

Resolve C1 and C2:

C1: $\neg S \vee M$

C2: S

M

So:

C4: M

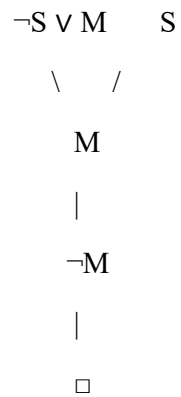
Now resolve C4 with C3:

C4: M

C3: $\neg M$

□

Resolution Tree



Conclusion

The Empty Clause □ is obtained.

Therefore, Rahul receives internal marks.

Question 3 – Library Membership

1. Propositional Logic

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Knowledge Base:

1. If Priya is a library member, then Priya can borrow books:
 $L \rightarrow B$
2. Priya is a library member:
L

Goal:

B

2. CNF Clauses

Convert the implication:

$L \rightarrow B$

to:

$\neg L \vee B$

Therefore:

- **C1: $\neg L \vee B$**
- **C2: L**

3. Negate the Goal

Goal:

B

Negated goal:

$\neg B$

- **C3: $\neg B$**

4. Resolution Steps

First, resolve C1 and C2:

C1: **$\neg L \vee B$**

C2: **L**

B

New clause:

C4: B

Now resolve C4 and C3:

C4: B

C3: $\neg B$

□

5. Resolution Tree

$\neg L \vee B$ L

\ /

B

|

$\neg B$

|

□

Final Conclusion

The Empty Clause □ is derived.

Therefore, Priya can borrow books.

Question 4 – Placement Eligibility

1. Propositional Logic

Let:

- A = Arun cleared the aptitude test.
- E = Arun is eligible for placement.

Knowledge Base:

1. If Arun clears the aptitude test, then Arun is eligible for placement:
 $A \rightarrow E$
2. Arun cleared the aptitude test:
 A

Goal:

E

2. CNF Clauses

Convert:

$A \rightarrow E$

to:

$\neg A \vee E$

Therefore:

- $C1: \neg A \vee E$
- $C2: A$

3. Negate the Goal

Goal:

E

Negated goal:

$\neg E$

- $C3: \neg E$

4. Resolution Step 1

Resolve:

$C1: \neg A \vee E$

with:

C2: A

The complementary literals are $\neg A$ and A .

Therefore:

C4: E

5. Resolution Step 2

Resolve:

C4: E

with:

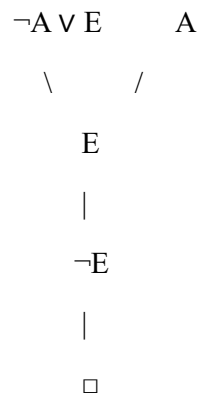
C3: $\neg E$

The complementary literals are E and $\neg E$.

Therefore:

C5: $\square$

Resolution Tree



Final Conclusion

Since the Empty Clause $\square$ is obtained, the negation of the goal is contradictory.

Therefore, Arun is eligible for placement.

Question 5 – Access Control System

1. Propositional Logic

Let:

- **P** = The user entered the correct password.
- **A** = The user is authenticated.
- **G** = The user is granted access.

Knowledge Base:

1. If the user enters the correct password, then the user is authenticated:
 $P \rightarrow A$
2. If the user is authenticated, then the user is granted access:
 $A \rightarrow G$
3. The user entered the correct password:
P

Goal:

G

2. Convert All Implications into CNF

First implication:

$$P \rightarrow A$$

becomes:

$$\neg P \vee A$$

Second implication:

$$A \rightarrow G$$

becomes:

$$\neg A \vee G$$

Fact:

$$P$$

Therefore, the clauses are:

- **C1:** $\neg P \vee A$
- **C2:** $\neg A \vee G$
- **C3:** **P**

3. Negate the Goal

Goal:

G

Negated goal:

$\neg G$

- **C4: $\neg G$**

4. Resolution Step 1

Resolve C1 and C3:

C1: $\neg P \vee A$

C3: P

A

Therefore:

C5: A

The user is authenticated.

5. Resolution Step 2

Resolve C2 and C5:

C2: $\neg A \vee G$

C5: A

G

Therefore:

C6: G

The user is granted access.

6. Resolution Step 3

Resolve C6 with the negated goal C4:

C6: G

C4: $\neg G$

□

The **Empty Clause** (□) is obtained.

Resolution Tree



Final Conclusion

Since the **Empty Clause** (□) is derived, the negated goal $\neg G$ is contradictory.

Therefore, the user is granted access.